

NP 2: 3SAT

NP-Completeness

1. Prove 3SAT is NP-complete
 - Cook-Levin Theorem (1971): SAT is NP-complete
 - Karp (1972): 21 other problems are NP-complete

3SAT

1. 3SAT
 - Input: Boolean formula f in CNF with n variables and m clauses where each clause has ≤ 3 literals
 - Output: Satisfying assignment if one exists and NO otherwise

Proof Outline

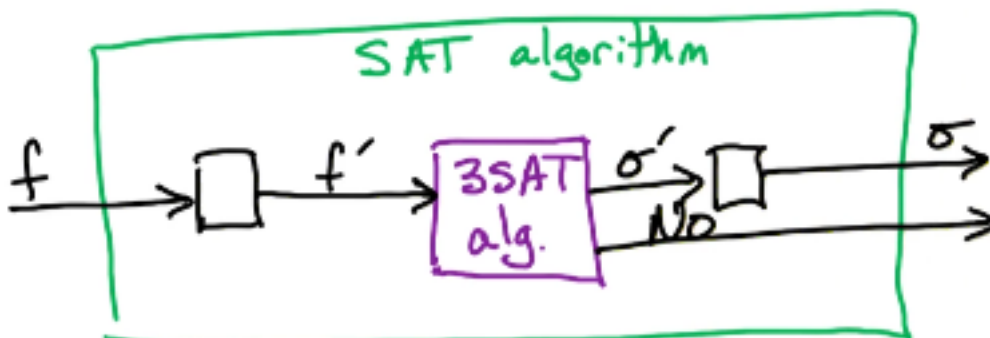
1. We'll show 3SAT is NP-complete
 - Need to show:
 - 3SAT is in NP
 - SAT $\rightarrow$ 3SAT
 - * Thus, for all A in NP, $A \rightarrow$ 3SAT

3SAT in NP

1. Given 3SAT input f and T/F assignment for $x_1, \dots, x_n$
 - For each clause C in f :
 - In $O(1)$ time can check that at least one literal in C is satisfied
 - $O(m)$ total time

SAT $\rightarrow$ 3SAT

1. Take input f for SAT
 - Need to create f' for 3SAT
 - f' has a satisfying assignment iff f has a satisfying assignment
 - f' has no satisfying assignment iff f has no satisfying assignment
 - Transform satisfying output σ' for $f' \rightarrow \sigma$ for f
 - σ' satisfies $f' \iff \sigma$ satisfies f



SAT $\rightarrow$ 3SAT

Example

1. $f = (x_3) \wedge (!x_2 \vee x_3 \vee !x_1 \vee !x_4) \wedge (x_2 \vee x_1)$
 - Input f' for 3SAT:
 - Keep C_1 and C_3 the same
 - For C_2 : Create a new variable y
 - * $C_2' = (!x_2 \vee x_3 \vee y) \wedge (!y \vee !x_1 \vee !x_4)$
 - Claim: C_2 is satisfiable iff C_2' is satisfiable

Claim: Forward

1. Proof of Claim:
 - $C_2 = (!x_2 \vee x_3 \vee !x_1 \vee !x_4)$
 - $C_2' = (!x_2 \vee x_3 \vee y) \wedge (!y \vee !x_1 \vee !x_4)$
 - Take satisfying assignment for C_2
 - if $x_2 = F$ or $x_3 = T$: then set $y = F$
 - if $x_1 = F$ or $x_4 = F$: then set $y = T$
 - In any case, C_2' is satisfied

Claim: Reverse

1. Proof of Claim:
 - $C_2 = (!x_2 \vee x_3 \vee !x_1 \vee !x_4)$
 - $C_2' = (!x_2 \vee x_3 \vee y) \wedge (!y \vee !x_1 \vee !x_4)$
 - Take satisfying assignment for C_2'
 - if $y = T$: then $x_1 = F$ or $x_4 = F$
 - if $y = F$: then $x_2 = F$ or $x_3 = T$
 - In any case, C_2 is satisfied

Quiz: 5-SAT 3-SAT

1. $C = (!x_2 \vee x_3 \vee !x_1 \vee !x_4 \vee x_5)$
 - Create two new variables y and z
 - Define C' where each clause is of size ≤ 3
 - C is satisfiable $\iff C'$ is satisfiable
2. $C' = (!x_2 \vee x_3 \vee y) \wedge (!y \vee !x_1 \vee z) \wedge (!z \vee !x_4 \vee x_5)$
 - $C \rightarrow C'$: $x_1 = F, y = T, z = F$
 - $C' \rightarrow C$:
 - if $y = F$ then x_2 or x_3 are satisfied
 - if $z = T$ then x_4 or x_5 are satisfied
 - $y = T$ and $z = F$ then x_1 is satisfied
3. For a clause of size k , we will create $k-3$ new variables and $k-2$ clauses

Big Clauses

1. $C = (a_1 \vee a_2 \vee \dots \vee a_k)$ where $a_1, a_2, \dots, a_k$ are literals
 - Create $k-3$ new variables $y_1, y_2, \dots, y_{k-3}$
 - Replace C by $k-2$ clauses
 - $C' = (a_1 \vee a_2 \vee y_1) \wedge \dots \wedge (!y_{k-4} \vee a_{k-2} \vee y_{k-3}) \wedge (!y_{k-3} \vee a_{k-1} \vee a_k)$
2. C is satisfiable $\iff C'$ is satisfiable

General Claim: Forward

1. Take assignment to $a_1, \dots, a_k$ satisfying C
 - Let a_i be min i where a_i is satisfied

- Since $a_i = T \rightarrow (i-1)^{\text{st}}$ clause of C' is satisfied
 - Set $y_1 = y_2 = \dots = y_{i-2} = T$ to satisfy 1st (i-2)
 - Set $y_{i-1} = y_i = \dots = y_{k-2} = F$ to satisfy rest

General Claim: Reverse

1. Take assignment to $a_1, \dots, a_k$ and $y_1, \dots, y_{k-3}$ satisfying C'
 - Need ≥ 1 is T
 - Suppose $a_1, \dots, a_k = F$
 - From clause 1 $\rightarrow y_1 = T$
 - From clause 2 $\rightarrow y_2 = T$
 - From clause $k-3 \rightarrow y_{k-3} = T$
 - * This means the final clause is not satisfied, so our assumption that all of our literals are set to false is incorrect

SAT 3SAT

1. Consider f for SAT
 - Create input f' for 3SAT:
 - For each clause C in f :
 - * if $|C| \leq 3$ then add C to f'
 - * if $|C| > 3$ then create $k-3$ new variables and add C' as defined before
 - f is satisfiable $\Leftrightarrow f'$ is satisfiable

3SAT Correctness

1. Given assignment to $x_1, \dots, x_n$ satisfying f
 - for clause C in f
 - There is an assignment to $k-3$ new variables so that C' is satisfied
2. Given satisfying assignment for f'
 - for C' in $f' \geq 1$ literal in C is satisfied
3. f is satisfiable $\Leftrightarrow f'$ is satisfiable

Satisfying Assignment

1. Consider f for SAT
2. Create input f' for 3SAT:
 - For each clause C in f :
 - if $|C| \leq 3$ then add C to f'
 - if $|C| > 3$ then
 - * create $k-3$ new variables and add C' as defined before
3. f is satisfiable $\Leftrightarrow f'$ is satisfiable
4. Satisfying assignment o' for $f' \rightarrow$ ignore new variables to get o
5. Runtime
 - f has n variables, m clauses
 - f' has $O(nm)$ variables and $O(nm)$ clauses
 - Polynomial

Practice Problems

1. DPV 8.3: Stingy SAT
2. DPV 8.8: Exact 4-SAT